Day 9 Study Notes

Sat 21 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Current electricity (part 2) (75 min) · Maths: Sequences and series (part 1) (75 min) · Chemistry: General organic chemistry (part 1) (60 min)
Exam	1 h 15 min	The Day 9 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Current electricity (part 2)

Yesterday covered resistance and power. Today finishes the chapter: cells, Kirchhoff's laws and the two bridges. The potentiometer is no longer in the syllabus, so skip it.

1. Cells, emf and internal resistance

- The emf ε is the potential difference across the cell when no current flows. Internal resistance r is the resistance of the cell itself.
- Current through an external resistance R: $I = \dfrac{\varepsilon}{R + r}$.
- **Discharging** (the cell drives the current): terminal voltage $V = \varepsilon - Ir$, which is less than the emf.
- **Charging** (current is forced in through the positive terminal): $V = \varepsilon + Ir$, which is greater than the emf.
- Short circuit ($R = 0$): $I = \dfrac{\varepsilon}{r}$ and $V = 0$. Open circuit: $I = 0$ and $V = \varepsilon$.
- **Maximum power** is delivered to the external resistance when $R = r$. Then $P_{max} = \dfrac{\varepsilon^2}{4r}$ and the efficiency is only 50%.

2. Combinations of cells

	Equivalent emf	Equivalent internal resistance
Series (helping)	$\varepsilon_1 + \varepsilon_2$	$r_1 + r_2$
Series (opposing)	$\varepsilon_1 - \varepsilon_2$	$r_1 + r_2$
Parallel	$\dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$	$\dfrac{r_1 r_2}{r_1 + r_2}$
n identical in series	$n\varepsilon$	nr
n identical in parallel	ε	$\dfrac{r}{n}$

Use cells in series when the external resistance is large compared with r, and in parallel when it is small.

3. Kirchhoff's laws

- **Junction rule:** the currents entering a junction add up to the currents leaving it. This is conservation of **charge**.
- **Loop rule:** the potential changes around any closed loop add up to zero. This is conservation of **energy**.
- Sign convention while walking round a loop: crossing a resistor in the direction of the current is a drop of IR; against the current it is a rise of IR. Crossing a cell from its negative to its positive terminal is a rise of ε.
- Method: mark a current in each branch, use the junction rule to cut down the unknowns, then write one loop equation for each remaining unknown. A negative answer only means the current flows opposite to the direction you marked.

4. Wheatstone bridge

- Four resistors P, Q, R, S form a quadrilateral, with a galvanometer across one diagonal and a battery across the other.
- **Balance condition:** $\dfrac{P}{Q} = \dfrac{R}{S}$. The galvanometer then carries no current.

- At balance the galvanometer branch can be removed. The bridge becomes $(P+Q)$ in parallel with $(R+S)$.
- The balance does not change if the battery and the galvanometer are exchanged, and it does not depend on the emf of the battery.
- The bridge is most sensitive when all four resistances are of the same order.

5. Metre bridge

- A Wheatstone bridge in which a uniform wire of length 100 cm supplies two of the arms. With a known resistance R in the left gap, an unknown X in the right gap and the balance point at l cm from the left end: $\dfrac{R}{X} = \dfrac{l}{100-l}$.
- Exchanging R and X moves the balance point from l to $100-l$.
- The error is least when the balance point is near the middle of the wire.

> **Solved example 1.** A cell of emf 6 V and internal resistance 1 Ω is connected to a 5 Ω resistor. Find the terminal voltage.
> $I = \dfrac{6}{5+1} = 1$ A. $V = \varepsilon - Ir = 6 - 1 = 5$ V. The same answer comes from $V = IR = 1 \times 5$.

> **Solved example 2.** In a metre bridge, a 6 Ω resistor is in the left gap and the balance point is 30 cm from the left end. Find the resistance in the right gap.
> $\dfrac{6}{X} = \dfrac{30}{70}$, so $X = 14$ Ω.

Traps. Terminal voltage is below the emf only while the cell is discharging. In parallel, identical cells do not add their emfs. In the metre bridge, check which gap holds the unknown before writing the ratio. Maximum power transfer needs $R = r$, and maximum current needs $R = 0$; these are different questions.

Mathematics · Sequences and series (part 1)

Sequences and series gives one to two questions per paper. Today covers the AP and the GP. Tomorrow's part covers means, the AM–GM inequality and special series.

1. Arithmetic progression (AP)

- First term a, common difference d. nth term: $a_n = a + (n-1)d$.
- Sum of n terms: $S_n = \dfrac{n}{2}\,[2a + (n-1)d] = \dfrac{n}{2}\,(a + l)$, where l is the last term.
- Number of terms from a to l: $n = \dfrac{l - a}{d} + 1$.
- From a sum to a term: $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$.
- If $S_n = An^2 + Bn$, the sequence is an AP with $d = 2A$ and $a_n = A(2n-1) + B$.
- a, b, c are in AP exactly when $2b = a + c$.
- Terms equidistant from the two ends have the same sum: $a_k + a_{n+1-k} = a_1 + a_n$.
- **Choosing terms:** three terms in AP as $a - d,\ a,\ a + d$; four terms as $a - 3d,\ a - d,\ a + d,\ a + 3d$. The sum then gives a directly.

2. Geometric progression (GP)

- First term a, common ratio r. nth term: $a_n = ar^{\,n-1}$.
- Sum of n terms: $S_n = \dfrac{a(r^n - 1)}{r - 1}$ for $r \neq 1$, and $S_n = na$ for $r = 1$.
- Sum to infinity: $S_\infty = \dfrac{a}{1 - r}$, only when $|r| < 1$.
- a, b, c are in GP exactly when $b^2 = ac$.
- **Choosing terms:** three terms in GP as $\dfrac{a}{r},\ a,\ ar$. The product then gives a^3 directly.

3. Properties that save time

- Terms of an AP picked at equal gaps (such as the 4th, 10th and 16th) are again in AP. Terms of a GP picked at equal gaps are again in GP.
- Adding or multiplying every term of an AP by a constant gives an AP. Multiplying every term of a GP by a constant, or raising every term to the same power, gives a GP.
- If $a_1, a_2, \ldots$ is a GP of positive terms, then $\log a_1, \log a_2, \ldots$ is an AP.
- For an AP, S_n is a quadratic in n with no constant term, and a_n is linear in n.

> **Solved example 1.** The sum of n terms of a sequence is $S_n = 2n^2 + 3n$. Find the 5th term.
> $a_5 = S_5 - S_4 = (50 + 15) - (32 + 12) = 65 - 44 = 21$. Check with the shortcut: $A(2n-1) + B = 2 \times 9 + 3 = 21$.

> **Solved example 2.** Three numbers in GP have sum 26 and product 216. Find them.
> Take $\dfrac{a}{r},\ a,\ ar$. The product gives $a^3 = 216$, so $a = 6$. Then $\dfrac{6}{r} + 6 + 6r = 26$, so $3r^2 - 10r + 3 = 0$ and $r = 3$ or $\dfrac{1}{3}$. The numbers are 2, 6, 18.

Traps. "Between 100 and 200" excludes both ends. S_∞ exists only for $|r| < 1$. $a_n = S_n - S_{n-1}$ does not hold for $n = 1$. When counting terms, add 1 after dividing the gap by d.

Chemistry · General organic chemistry (part 1)

General organic chemistry is the base for every organic chapter that follows. It gives one direct question in most papers and decides many others on acidity, basicity and stability of intermediates.

1. Hybridisation of carbon

Carbon has	Hybridisation	Shape	Bond angle	s character
4 σ bonds	sp^3	tetrahedral	109.5°	25%
3 σ bonds, 1 π bond	sp^2	trigonal planar	120°	33%
2 σ bonds, 2 π bonds	sp	linear	180°	50%

- Count σ bonds on the carbon: 4 means sp^3, 3 means sp^2, 2 means sp.
- Every single bond is one σ. A double bond is one σ and one π. A triple bond is one σ and two π.
- More s character means a more electronegative carbon, a shorter bond and a more acidic C–H. Acidity: ethyne > ethene > ethane.
- The middle carbon of an allene (C=C=C) is sp. The carbon of a carbocation is sp^2.

2. Bond fission and reagents

- **Homolytic fission:** each atom keeps one electron. It gives **free radicals**. Favoured by heat, light and peroxides, in non-polar conditions.
- **Heterolytic fission:** one atom keeps both electrons. It gives a **carbocation** and an anion, or a **carbanion** and a cation. Favoured by polar solvents.
- **Electrophiles** are electron-seeking species: H^+, NO_2^+, Br^+, carbocations, and neutral molecules with an incomplete octet such as BF_3, $AlCl_3$ and SO_3.
- **Nucleophiles** are electron-rich species with a lone pair or a negative charge: OH^-, CN^-, NH_3, H_2O, ROH.

3. Inductive effect

- A permanent shift of σ electrons along a chain. It dies out quickly, and is negligible after the third carbon.
- **−I groups** (pull electrons): $–NO_2$ > –CN > –COOH > –F > –Cl > –Br > –I > –OH > $–C_6H_5$.
- **+I groups** (push electrons): alkyl groups, with $(CH_3)_3C–$ > $(CH_3)_2CH–$ > $CH_3CH_2–$ > $CH_3–$.
- −I groups make an acid stronger, more so when there are more of them and when they are closer to –COOH. Cl_3CCOOH > $Cl_2CHCOOH$ > $ClCH_2COOH$ > CH_3COOH. Also HCOOH > CH_3COOH, because the methyl group is +I.

4. Resonance

- Resonance structures differ only in the positions of π electrons and lone pairs. The atoms do not move.
- The real molecule is the resonance hybrid. It is more stable than any single structure, by the resonance energy.
- A structure contributes more when it has more covalent bonds, complete octets, less charge separation, and a negative charge on the more electronegative atom.
- **+R groups** (give a lone pair to the π system): –OH, –OR, $–NH_2$, –NHR, –X (halogens).

- **−R groups** (pull π electrons out): $–NO_2$, –CN, –CHO, –COR, –COOH.
- Halogens are −I and +R. In reactions of the benzene ring the −I effect controls the rate, and the +R effect controls the position.

5. Electromeric effect

- A complete transfer of a shared π pair to one atom, only when an attacking reagent is present. It is temporary, and it reverses when the reagent is removed.
- +E: the π electrons move to the atom that the reagent joins, as in H^+ adding to an alkene. −E: they move away from it, as in CN^- adding to a carbonyl group.

6. Hyperconjugation

- Delocalisation of the σ electrons of an α C–H bond into an adjacent empty p orbital, a half-filled p orbital or a π bond. It is permanent, and is also called no-bond resonance.
- Number of hyperconjugative structures = number of **α hydrogens**, the hydrogens on carbons directly attached to the positive carbon, the radical carbon or the double bond.
- Carbocation stability: $(CH_3)_3C^+$ (9 α-H) > $(CH_3)_2CH^+$ (6) > $CH_3CH_2^+$ (3) > CH_3^+ (0). Free radicals follow the same order.
- Alkene stability also rises with the number of α hydrogens, so more substituted alkenes are more stable.

> **Solved example 1.** Count the σ and π bonds in $CH_3–CH=CH_2$.
> C–H bonds: 3 + 1 + 2 = 6. C–C σ bonds: 2. Total σ = 8. The double bond gives 1 π.

> **Solved example 2.** How many hyperconjugative structures does $CH_3–CH=CH_2$ have?
> Only the CH_3 carbon is attached to the double bond, and it carries 3 hydrogens. So there are 3.

Traps. Count the C–H bonds when counting σ bonds; they are not drawn in condensed formulae. –OH and $–NH_2$ are −I but +R, and the +R effect usually wins. Hydrogens on the positive carbon itself are not α hydrogens. BF_3 is neutral but is an electrophile.